

Course: B. Tech.

Branch : Mechanical

Semester : III

Subject Code & Name: BTMC302 , Fluid Mechanics

Max Marks: 60

Date: 10/08/2023

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) What are the conditions of equilibrium of floating body and sub-merged Body?	CO1	6
B) Define and explain the following fluid properties. Viscosity, specific gravity, specific volume, specific weight, surface tension, compressibility.	CO1	6
C) The space between two square flat parallel plates is filled with oil. Each side of the plate is 60 cm. The thickness of the oil film is 12.5 mm. The upper plate, which moves at 2.5 metre per sec requires a force of 98.1 N to maintain the speed. Determine : (i) the dynamic viscosity of the oil , and (ii) the kinematic viscosity of the oil if the specific gravity of the oil is 0.95	CO1	6
Q.2 Solve Any Two of the following.		12
A) Define the following flow: (i) Steady Flow (ii) Non-Uniform Flow (iii) Laminar Flow (iv) Turbulent Flow (v) Compressible Flow (vi) Irrotational Flow	CO2	6
B) Derive the general equation for continuity for a three dimensional flow in Cartesian Co-ordinates for a steady incompressible flow	CO2	6
C) The velocity potential function is given by $\phi = 5(x^2 - y^2)$. Calculate the velocity components at the point (4, 5).	CO2	6

Q. 3	Solve Any Two of the following.		12
A)	Derive an expression for the velocity distribution for laminar flow through a circular pipe. Also sketch the velocity distribution and shear stress distribution across a section of the pipe	CO3	6
B)	What is minor energy loss in the pipe? Explain various types of minor losses in pipe flow.	CO3	6
C)	Calculate the discharge through a pipe of diameter 200 mm when the difference of pressure head between the two ends of a pipe 500 m apart is 4 m of water. Take the value of $f = 0.009$ in the formula $h_f = \frac{4fLV^2}{2g*d}$	CO3	6
Q.4	Solve Any Two of the following.		12
A)	State Buckingham's π -Theorem and explain procedure for determining the π -groups and their functional relationship	CO4	6
B)	What is Siphon? Explain its working with neat sketch.	CO4	6
C)	Derive Dupit's equation & Explain concept of equivalent pipe.	CO4	6
Q. 5	Solve Any Two of the following.		12
A)	Define a centrifugal pump. Explain the working of a single-stage centrifugal pump with sketches.	CO5	6
B)	What is priming? Why is it necessary?	CO5	6
C)	Write a short note on multistage centrifugal pump	CO5	6

***** End *****